Geometry-Conditioned Appearance Retargeting for Image-to-3D Generation

Anonymous CVPR submission

Abstract

Recent image-to-3D models generate high-quality assets, but modifying geometry while preserving its identity remains challenging. A key difficulty is that the target geometry and original source views remains different, making direct feature fusion unreliable. We present GEAR3D, a geometry-conditioned image-to-3D framework that leverages VLM-guided editing to establish a geometry-aligned target view and appearance retargeting to adapt source evidence according to the deformed target view. By selectively adapting source features within projection-based conditioning, GEAR3D preserves original appearance while following the intended geometry. We further introduce DeformNeXt-10k, a paired deformation dataset with surface correspondences and multi-view observations. Experiments demonstrate improved source appearance preservation and target alignment, enabling flexible and high-fidelity 3D editing.

1. Introduction

Recent image-to-3D models have made it possible to generate high-quality 3D assets from a single [16, 20, 22, 36, 37, 46] or multiple views [3, 15, 26, 27, 38], simplifying 3D content creation and facilitating product visualization or virtual prototyping. Their goal is to reconstruct or generate 3D assets whose geometry and appearance are consistent with the input observations.

However, practical asset creation often requires more than mere reconstruction. For example, a designer may want to modify the proportions of furniture while preserving its distinctive upholstery, or a user may want to reshape a product while retaining its original texture and material. Given multi-view images of a source object and a user-specified target geometry, our goal is to generate a high-quality 3D asset that follows the target shape while retaining the source-specific appearance. We formulate this problem as *geometry-conditioned appearance retargeting* in image-to-3D generation process. It enables appearance reuse across geometric variations without requiring manual texture editing, which can accelerate 3D asset prototyping and also production process.

Despite its practical relevance, this problem presents a challenge distinct from conventional image-to-3D reconstruction. Existing approaches only partially address the requirements of geometry-guided editing. Generating edited multi-view images [14, 28, 31, 45] can introduce cross-view inconsistencies, while directly deforming a source mesh may produce unintended distortions or artifacts [5, 42]. Likewise, modifying geometric latents or applying spatial controls [9] may change the geometry without ensuring that the visual condition reflects the edited shape. We therefore employ a geometry-guided vision-language model (VLM) to transform the original target-view image according to a user-specified geometric condition. By leveraging its broad semantic and visual priors, this step expresses diverse geometric modifications as an image-domain target condition compatible with pretrained image-to-3D generation.

However, a geometry-aligned target view alone cannot preserve the object-specific textures and materials, particularly in regions not visible from the target viewpoint. Therefore, we incorporate complementary appearance evidence from the original multi-view source observations. While existing image-to-3D models generally assume that the input observations correspond to the geometry being reconstructed, the edited target and original source images in our setting describe different shapes. Consequently, directly projecting source image features onto the target geometry may associate unrelated surface regions with appearance cues, leading to appearance artifacts and quality degradation. Existing source-aware editing methods [11] demonstrate the importance of preserving source appearance, but do not explicitly address this cross-state feature alignment within the 3D generation process. The key challenge is therefore not simply to impose a new geometry, but to *transfer and adapt appearance evidence to the target geometry.*

To this end, we propose *GEAR3D*, a new framework for ***ge****ometry-conditioned* ***a****ppearance* ***r****etargeting* in image-to-**3D** generation that explicitly adapts source-state appearance to the target geometry. Our framework uses the geometry-aligned target view as a structural anchor and incorporates complementary appearance evidence from the original source views. Specifically, we introduce three core modules: (1) *semantic feature matching* to establish correspondences across geometric deformation, (2) *matching-aware feature aggregation* to selectively aggregate source

appearance features using ray-shared matching scores, and (3) *target-aware feature retargeting* to adapt the retrieved features to the target state through residual source-feature correction. Furthermore, we construct *DeformNeXt-10k*, a paired deformation dataset with preserved surface correspondences and multi-view observations, to supervise geometry-conditioned appearance retargeting.

Our main contributions are summarized as follows:

- We formulate *geometry-conditioned appearance retargeting* to adapt source appearance to the target geometry.
- We propose a target-anchored 3D generative framework combining *semantic feature matching*, *matching-aware feature aggregation*, and *target-aware feature retargeting* to align, retrieve, and adapt source appearance evidence.
- We construct *DeformNeXt-10k*, a paired deformation dataset with surface correspondences and multi-view observations, for systematic training and evaluation.

2. Related Work

2.1. Image-Conditioned 3D Generation

Image-conditioned 3D generation has advanced through two complementary paradigms. Feed-forward reconstruction methods directly infer 3D geometry and appearance from input images using learned geometric priors [1, 13, 35, 40]. In contrast, generative approaches learn structured or native 3D latent representations to synthesize high-fidelity assets with complex geometry and appearance [21, 22, 36, 37, 46]. Recent studies further improve fidelity to input observations through pixel-aligned conditioning and reconstruction-guided generation [3, 15, 20]. Among these, Pixal3D [20] incorporates projected image features into 3D generation to better preserve observed appearance. However, these methods primarily assume that the conditioning images depict the geometry to be generated. In contrast, our framework combines a geometry-aligned target view with original source views depicting different geometric states, explicitly retargeting source appearance evidence across geometric variations.

2.2. Geometry-Guided 3D Editing

Geometry-guided 3D editing aims to modify existing 3D assets while preserving their structural plausibility and visual cues. Prior methods achieve geometric modifications through deformation constraints, learned shape priors, and image- or text-guided optimization [5, 6, 18, 42]. More recent approaches incorporate spatial controls into generative frameworks or introduce dedicated 3D editing models [8, 9]. However, geometric controls alone may leave the visual conditioning inconsistent with the edited shape. Complementary to geometry editing, reference-guided texturing methods synthesize consistent appearance on prescribed geometry through diffusion-based view generation and texture transfer [2, 4, 39, 41]. Specifically, ShapeUP [11] performs geometry editing through latent-space translation and subsequently guides texture synthesis using the edited geometry and renderings of the source asset. While ShapeUP demonstrates the importance of preserving source appearance during geometric modifications, our approach uses a VLM-edited target view as a structural anchor and explicitly retargets source-view features across geometric states within the 3D generation process.

2.3. Correspondence-Guided Feature Fusion

Dense correspondence estimation has benefited from semantic representations learned by diffusion and self-supervised vision models [24, 29, 30, 43]. Beyond semantic descriptors, learned matching frameworks and geometric foundation models provide complementary structural and 3D-aware priors for establishing correspondences [7, 19, 32, 33]. In particular, RoMa-Ω [25] investigates the correspondence capabilities of VGGT-Ω [33] features and incorporates selected geometric representations into a learned dense matching framework. Its findings suggest that geometrically informative features do not necessarily provide reliable correspondences through direct feature similarity, highlighting the importance of dedicated feature matching.

Correspondence and geometric alignment have also been explored for consistent multi-view generation. Existing methods employ synchronized denoising, cross-view attention, and geometry-aware feature aggregation to maintain consistency across generated views [12, 14, 17, 23, 28, 31, 45]. While these approaches primarily focus on correspondence estimation or cross-view consistency, our work leverages semantic correspondences for a different purpose: retrieving source-specific appearance information across geometric state changes and adapting the matched features to guide target-state 3D generation.

3. Method

3.1. Preliminaries: Projection Injection

Unlike attention-based 3D generators such as TRELLIS [36], which implicitly learn associations between image tokens and 3D queries, Pixal3D [20] explicitly establishes 2D-3D spatial correspondence through back-projection, enabling high-fidelity image-to-3D generation. We adopt this projection-based conditioning principle, as its explicit spatial associations provide a suitable foundation for retrieving and adapting source appearance evidence across geometric changes. Given multi-view images $\{I_v\}_{v=1}^{V}$, the projected features at voxel $\mathbf{x}$ are aggregated as

$$\mathbf{F}^{3D}(\mathbf{x}) = \frac{1}{V}\sum_{v=1}^{V} \mathrm{Sample}\left(\Phi(I_v), \pi_v(\mathbf{x})\right), \qquad (1)$$

where Φ denotes the image encoder, π_v is the camera projection for view v, and Sample denotes bilinear sampling. However, explicit geometric projection alone does not guarantee semantic consistency when the input views depict different shapes. Consequently, features projected onto the same 3D query may originate from semantically unrelated object regions, leading to appearance misalignment. We therefore preserve the target-view conditioning while adapting source appearance through semantic correspondence-guided feature aggregation and retargeting.

3.2. Data Construction: DeformNeXt-10k

[TODO: Add example figure.] To supervise appearance retargeting across geometric variations, we construct *DeformNeXt-10k* from textured 3D assets and hierarchical part annotations of PartNeXt [34]. For each source object, we employ donor-guided GeoStyle [5] deformation with part-aware assembly constraints to generate diverse geometric variations. The resulting candidates are screened for meaningful deformations and major geometric artifacts. Importantly, each source mesh $M_0 = (V_0, F, A_0)$ and its deformed counterpart $M_1 = (V_1, F, A_0)$ share identical face connectivity and appearance attributes, including UV coordinates and materials. This preserves exact surface correspondences through shared face indices and barycentric coordinates while retaining the original appearance. We render both geometric states from calibrated viewpoints to obtain paired target and source observations for training. The resulting dataset contains 10,165 paired deformations, providing correspondence supervision for semantic feature matching and supervised training of geometry-conditioned appearance retargeting.

3.3. VLM-Guided Target-View Editing

[TODO: Add example figure.] Direct mesh-based editing often requires explicit deformation constraints or optimization [5, 42], while spatial geometry controls [9] may leave the original visual condition inconsistent with the edited shape. These limitations make it challenging to achieve flexible geometric editing while maintaining geometry-image consistency. To address this, we employ a vision-language model (VLM) to translate user-specified geometric modifications into a geometry-aligned target image. Given an original target-view image I_t^0 and a desired geometric condition $\mathcal{G}^\star$, we obtain

$$I_t^{\text{edit}} = \mathcal{E}_{\text{VLM}}\left(I_t^0, \mathcal{G}^\star\right), \tag{2}$$

where $\mathcal{E}_{\text{VLM}}$ denotes the geometry-guided image editing process. By leveraging the real-world commonsense knowledge of VLMs, this approach translates diverse geometry conditions into compatible edited target images. The edited target view serves as a structural anchor, while the original source views retain complementary appearance cues. However, these observations have different geometry, requiring explicit appearance retargeting during 3D generation.

3.4. GEAR3D Framework

[TODO: Add example figure.] Given the geometry-aligned target view and a set of original source views indexed by $\mathcal{S}$, *GEAR3D* adapts source appearance evidence to the target geometric state within a pretrained projection-based 3D generator. Let $V = |\mathcal{S}| + 1$ denote the total number of conditioning views, including the target view. Our framework consists of three components: semantic feature matching, matching-aware feature aggregation, and target-aware feature retargeting.

Semantic Feature Matching. To transfer source appearance across geometric variations, we first establish semantic correspondences between the geometry-aligned target view and the original source views. We employ geometry-aware descriptors from a frozen VGGT-Ω [33] backbone for correspondence estimation. For each target patch, we retrieve the top-K source patch candidates based on feature similarity. However, direct feature similarity may produce unreliable correspondences under deformation and pose variations. To improve matching accuracy, we adapt a lightweight matching decoder [25] that contextualizes target descriptors through cross-attention and refines candidate matching scores. For each target patch m, let $\mathcal{C}_m$ denote its retrieved source candidates and s_{mk} the refined matching score for candidate k. Since a target region may correspond to multiple visible source patches across views, we construct a set of valid positive matches $\mathcal{P}_m$ using the preserved surface correspondences in *DeformNeXt-10k*. We optimize the decoder with a multi-positive matching objective:

$$\mathcal{L}_{\text{match}} = -\log \frac{\sum_{k \in \mathcal{C}_m \cap \mathcal{P}_m} \exp(s_{mk})}{\sum_{k \in \mathcal{C}_m} \exp(s_{mk})}. \tag{3}$$

Only target patches with valid positive candidates contribute to training, and the matcher remains frozen during 3D generation.

Matching-Aware Feature Aggregation. **[TODO: Add example figure.]** In projection-based conditioning, a source image patch provides the same feature to multiple 3D queries along its camera ray. However, these queries may project onto different target patches, making their cross-geometry correspondences ambiguous. Therefore, independently matching each query yield inconsistent source evidence, while selecting a single best match risks missing valid correspondences. To address this, we aggregate complementary matching proposals from target patches associated with the same source-camera ray. Specifically, for each

source view $j \in \mathcal{S}$, we group voxels projecting onto the same source patch p into a ray group $r = (j, p)$. We denote the distinct target patches observed through these voxels by $\mathcal{M}_r$. For each target patch $m \in \mathcal{M}_r$, the matching decoder produces scores s_{mk} over its source candidates $\mathcal{C}_m$, which are normalized as

$$p_m(k) = \frac{\exp(s_{mk}/\tau)}{\sum_{\ell \in \mathcal{C}_m} \exp(s_{m\ell}/\tau)}, \tag{4}$$

where τ is the softmax temperature and $p_m(k) = 0$ for candidates outside $\mathcal{C}_m$. We aggregate these matching distributions and retain the top-K candidates:

$$\bar{p}_r(k) = \frac{1}{|\mathcal{M}_r|} \sum_{m \in \mathcal{M}_r} p_m(k), \qquad \mathcal{K}_r = \mathrm{TopK}(\bar{p}_r). \tag{5}$$

The retained scores are renormalized as $p_r(k) = \bar{p}_r(k) / \sum_{\ell \in \mathcal{K}_r} \bar{p}_r(\ell)$. The aggregated source evidence is then computed as

$$\mathbf{E}_r = \sum_{k \in \mathcal{K}_r} p_r(k) \mathbf{v}_k, \tag{6}$$

where $\mathbf{v}_k$ denotes the native DINOv3 feature used to condition the pretrained 3D generator, rather than the VGGT-Ω descriptors used by the separately trained matching decoder. By combining matching probabilities from multiple target patches, we aggregate relevant source features rather than relying on a single correspondence. The aggregated feature is shared across 3D queries projecting onto the same source patch and subsequently adapted to the target geometry through target-aware feature retargeting.

Target-Aware Feature Retargeting. Although matching-aware aggregation retrieves relevant source appearance evidence, directly replacing the original source features may disrupt the pretrained conditioning. We therefore adapt the existing source features toward the retrieved evidence, conditioned on the target geometry. For each ray r, let $\mathbf{S}_r$ denote the original source feature and $\mathbf{T}_r$ the target-view context pooled over the corresponding 3D queries. We compute the retargeted source feature as

$$\begin{aligned} \mathbf{R}_r &= \mathcal{R}_\phi\left([\mathbf{E}_r - \mathbf{S}_r, \mathbf{T}_r]\right), \\ \widetilde{\mathbf{S}}_r &= \mathbf{S}_r + u_r\left[g(\mathbf{E}_r - \mathbf{S}_r) + \mathbf{R}_r\right], \end{aligned} \tag{7}$$

where $\mathcal{R}_\phi$ is a learnable retargeting network, u_r controls the use of matched evidence, and g is a bounded, signed gate. The feature difference $\mathbf{E}_r - \mathbf{S}_r$ guides the source adaptation, while $\mathbf{T}_r$ provides target-specific context. Rather than introducing an additional conditioning stream, we update the existing source evidence through a residual formulation:

$$\mathbf{C} = \mathbf{C}_{\mathrm{MV}} + \frac{1}{V} \sum_{j \in \mathcal{S}} \mathcal{P}_j(\widetilde{\mathbf{S}}_j - \mathbf{S}_j), \tag{8}$$

where $\mathcal{P}_j$ projects source-view features into the shared 3D space. This formulation preserves the pretrained conditioning at initialization while allowing source features to be selectively adapted to the target geometry. Rays without valid matching evidence retain their original source features.

4. Experiments

4.1. Experimental Setting

Dataset. We conduct experiments on DeformNeXt-10k, comprising 10,165 paired deformations from 3,483 source objects across 50 categories. The dataset is divided into 9,149 training, 508 validation, and 508 test examples, with source identities and geometry clusters disjoint across splits. Each example provides multi-view observations of the original source object and its deformed counterpart, with preserved surface correspondences and appearance. For controlled evaluation, we use rendered target-state images as geometry-aligned target observations, isolating appearance retargeting from variations in VLM-based editing. Further details are provided in the Supp. Mat..

Baselines. We compare against three categories of methods to evaluate different strategies for geometry-conditioned 3D editing. *Image-to-3D generation* methods, TRELLIS.2 [37] and Pixal3D [20], generate 3D assets from the edited target image alone, without access to the original source appearance. *Multi-view reconstruction* methods include ReconViaGen [3], which directly combines edited target and original source views, and MVAdapter [14] and CorrAdapter [45], which generate edited multi-view images followed by a common ReconViaGen reconstruction backbone. *Geometry-guided editing* methods include ShapeUP [11], which edits an existing source mesh, and SpaceControl [9], which uses a predicted target mesh as spatial control with original source-image appearance conditioning. Together, these baselines examine the trade-offs between target-view reconstruction, multi-view conditioning, and explicit geometric control.

Metrics. We evaluate three aspects of 3D editing quality: source appearance preservation, target alignment, and overall visual quality. Appearance preservation is measured using LPIPS [44], DreamSim [10], and CLIP similarity, against reference renderings of the target geometry with the original source materials. Target alignment is evaluated using PSNR, SSIM, and mask IoU for rendered images, together with Chamfer Distance (CD) and surface F-Score for geometry. We additionally report FID and KID to assess the overall distribution of generated renderings.

Implementation Details. We extend the single-view Pixal3D [20], built on TRELLIS.2 with DINOv3 pro-

Table 1. **Quantitative comparison on test set of DeformNeXt-10k.** Best and second best results are highlighted.

Method	LPIPS↓	DreamSim↓	CLIP↑	FID↓	KID↓	PSNR↑	SSIM↑	M-IoU↑	CD↓	F-Score↑
TRELLIS.2 [37]	0.0942	0.2625	0.9028	61.21	3.18	18.01	0.9232	0.4575	0.0883	0.2969
Pixal3D [20]	0.1064	0.2671	0.9070	68.39	5.67	16.96	0.9162	0.4075	0.1028	0.1535
ReconViaGen [3]	**0.0810**	**0.1911**	**0.9232**	**45.92**	**0.53**	**19.12**	**0.9278**	**0.5873**	**0.0618**	0.3891
MVAdapter [14]	0.1036	0.2881	0.8905	66.56	4.32	17.35	0.9125	0.4729	0.0810	0.2359
CorrAdapter [45]	0.1030	0.2868	0.8914	65.68	4.31	17.37	0.9148	0.4658	0.0828	0.2317
ShapeUP [11]	0.0854	0.3021	0.8826	74.37	7.08	17.42	0.9217	0.5227	0.0778	**0.4577**
SpaceControl [9]	0.0969	0.2898	0.8975	64.78	3.72	16.76	0.9179	0.4608	0.0869	0.2955
MV-Pixal3D (FT)	–	–	–	–	–	–	–	–	–	–
Ours (Full)	–	–	–	–	–	–	–	–	–	–

jection conditioning, to multi-view generation and train the resulting model. We then fine-tune SS64 and PBR1024 on *DeformNeXt-10k* using LoRA, while keeping Shape512/1024 frozen. The matching decoder is separately trained with surface correspondence supervision and frozen alongside the image encoders during generative fine-tuning. SS64 and PBR1024 are trained for 5 epochs, with batch sizes of 96 and 24, respectively. Further details are provided in the Supp. Mat.

4.2. Generation Results

DeformNeXt-10k. We evaluate 3D editing performance on DeformNeXt-10k, comparing source appearance preservation and target geometry alignment against existing approaches. Table 1 presents results on 502 commonly successful test examples. Among existing methods, ReconViaGen achieves the best performance on nine of the ten metrics, including source LPIPS (0.0810), DreamSim (0.1911), and Chamfer Distance (0.0618), highlighting the value of incorporating original source observations. ShapeUP achieves the highest surface F-Score (0.4577), but exhibits lower source appearance fidelity, particularly in DreamSim (0.3021). Meanwhile, MVAdapter and CorrAdapter show lower geometric fidelity than direct multi-view reconstruction, suggesting that generating edited multi-view images does not necessarily ensure accurate geometry. These results highlight the challenge of jointly preserving source appearance and achieving target geometry alignment. **[TODO: Insert GEAR3D results evaluated under the same 502-example benchmark protocol. Report improvements over the strongest baselines in appearance preservation, target alignment, and geometry quality.] [TODO: Add a qualitative comparison figure showing the edited target, original source views, ground truth, representative baselines, and GEAR3D results across diverse geometric deformations.]**

ShapeUP Benchmark. [TODO: Concat table to side of Table 1] We additionally consider the ShapeUP benchmark to evaluate generalization beyond DeformNeXt-10k. **[TODO: Specify the benchmark composition and evaluation protocol. Report quantitative comparisons with ShapeUP and other applicable baselines, including geometry editing fidelity and source appearance preservation. Add a quantitative table and qualitative comparison figure.]**

User Study. To evaluate the practical effectiveness of our complete geometry-guided editing pipeline, we conduct a user study using VLM-generated target images from user-specified geometric conditions. Unlike the controlled benchmarks using rendered target-state images, this evaluation considers the full editing process, from geometric control to textured 3D generation. We compare *GEAR3D* with representative baselines across diverse geometric modifications, assessing adherence to the intended geometry, preservation of source appearance, and overall editing quality. **[TODO: Generate end-to-end editing results using actual geometry conditions and VLM-edited target images. Select representative baselines and construct a diverse evaluation set. Conduct a blinded user study, report participant preferences for geometry alignment, appearance preservation, and overall quality, and include qualitative comparisons and a user-study results table.]**

4.3. Ablation Study

Semantic Feature Matching. We examine whether learned semantic correspondence provides more informative source evidence across geometric changes. As shown in Table 2, the decoder and score refinement increase Exact@1 from 1.25% to 8.56% on the same candidate support. Combining proposals from distinct target observations further improves Exact@1 to 11.19% and Exact@64 coverage from 50.76% to 73.36%. The matching variants also show progressively lower surface distance and higher F-Score in Table 3.

Table 2. **Cross-state correspondence retrieval.** Exact@K scores are percentages over 50,493 observable relations. The first three variants re-rank candidates on mean-ray support; the final row combines proposals from distinct target patches.

Variant	Exact@1↑	Exact@8↑	Exact@64↑
Raw cosine	1.25	11.09	50.76
+ Matching decoder	7.73	28.47	50.76
+ Score refinement	8.56	29.92	50.76
+ **Ray fusion (Ours)**	11.19	45.68	73.36

Table 3. **Geometry-stage ablation.** The full model uses ray-fused, score-weighted Top-K evidence. Matching variants retain the same Fixed-style ray-level delivery. CD and F-Score are multiplied by 10^4 and 10^2, respectively, for display.

Variant	CD↓	F-Score↑
Ours (Ray-fused Top-K)	127.57	84.41
Semantic feature matching		
Spatial retrieval	129.70	83.93
Semantic retrieval	128.79	84.10
Decoder matching (cosine)	128.07	84.24
Matching-aware feature aggregation		
Top-1 selection	127.80	84.01
Uniform Top-K weighting	128.30	84.25
Target-aware feature retargeting		
Without retargeting	129.56	83.50

Table 4. **Appearance-stage ablation.** All variants share identical generated geometry and rendering conditions. LPIPS and DreamSim are multiplied by 10^3 for display. The Top-K variant uses a learned Top-K feature readout but retains the Top-1 carrier/source-use path.

Variant	LPIPS↓	DreamSim↓	CLIP↑
Original conditioning	53.51	158.35	0.9422
Top-1 retargeting (Ours)	53.29	157.83	0.9417
Top-K retargeting	53.36	158.26	0.9416

Matching-Aware Feature Aggregation. We examine whether retaining and weighting multiple matches is more effective than relying on a single correspondence. The full model, which includes ray-level proposal fusion and score-weighted Top-K aggregation, achieves the lowest CD and highest F-Score, outperforming Top-1 selection and uniform weighting in Table 3. Table 2 illustrates the additional candidate coverage of ray fusion; the geometry comparison evaluates aggregation alternatives under the selected ray-level delivery rather than separately removing ray fusion.

Target-Aware Feature Retargeting. Relevant source evidence must also be adapted to the target state. Removing residual retargeting increases scaled CD from 127.57 to 129.56 and lowers scaled F-Score from 84.41 to 83.50 (Table 3), supporting the value of target-aware adaptation beyond retrieval alone.

Appearance Retargeting. While geometry generation benefits from aggregating complementary correspondences, appearance synthesis requires precise source evidence to preserve fine-grained texture and material. Aggregating multiple source features may combine unrelated appearance cues and degrade visual fidelity. On identical generated geometry, Top-1 retargeting achieves the lowest LPIPS and DreamSim among the variants (Table 4). These results motivate stage-specific source adaptation: weighted Top-K aggregation for geometry and Top-1 retargeting for appearance. Additional results are provided in the Supp. Mat..

5. Conclusion

We presented *GEAR3D*, a geometry-conditioned image-to-3D framework that combines VLM-guided target-view editing with semantic feature matching, matching-aware feature aggregation, and target-aware feature retargeting. Experiments on *DeformNeXt-10k* demonstrate improved source appearance preservation and target alignment across geometric variations.

Limitation and Future Work. Our method relies on reliable target-view editing and semantic correspondences, which may become ambiguous under severe deformation or occlusion. Future work will extend appearance retargeting to topology-changing edits and unmatched regions, while exploring more diverse VLM-guided editing scenarios.

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Geometry-Conditioned Appearance Retargeting for Image-to-3D Generation

Supplementary Material

Overview

TODO

A. DeformNeXt-10k Construction Details

Source assets and canonical correspondence. We construct DeformNeXt-10k from textured PartNeXt assets and their hierarchical part annotations [34]. For each source mesh $M_0 = (V_0, F, A_0)$, we generate edited variants $M_{1,k} = (V_{1,k}, F, A_0)$, modifying vertex positions while preserving the ordered face array and source UV, material, and corner attributes. Consequently, corresponding points can be identified by the same face index and barycentric coordinates, without nearest-neighbor reassignment. This correspondence is used to construct training labels for cross-state matching; it is not assumed for surfaces decoded by the generative model.

Donor-guided deformation. We select same-category donor objects using fixed semantic-role and structural-compatibility heuristics. Roles are grouped semantically rather than requiring one-to-one correspondences between individual mesh components, allowing objects with different part counts and disconnected components. Eight renders of each donor with its original appearance are used to personalize a pretrained SDXL prior through GeoStyle [5]. Donor-specific DreamBooth LoRA is trained for 800 updates (rank 16, resolution 512, learning rate 10^{-4}), and its weights can be reused across source–donor pairs. We use the personalization caption "A photo of TOK sculpture" and the deformation prompt "A TOK style sculpture of a `<category>`". The donor supplies a morphology prior for proposing edits; neither the donor geometry nor its images are provided to the appearance-retargeting model at inference.

Part-aware assembly. GeoStyle proposes per-face Jacobian changes through image-space score distillation. To better preserve the source assembly, we deform selected primary surfaces while treating complementary physical components as rigid followers. Hierarchical semantic labels identify candidate roles, but do not provide ground-truth mechanical joints. We instead derive physical components from source connectivity and coincident-vertex classes, and estimate attachment constraints from nearby source surfaces using vertex-to-triangle patches, barycentric coordinates, and rest-frame offsets. The default attachment radius is 0.006 times the source diagonal, with at most 96 contact samples per component. Reconstruction alternates four projected Poisson/rigid steps, followed by a Dirichlet solve for follower positions. These constraints improve assembly coherence without guaranteeing collision-free geometry.

Optimization and optional refinement. The main production configuration performs 300 deformation iterations at resolution 512 with batch size one (seed 37), using main/auxiliary score-distillation weights of 2.5/0.25 and a cage-loss epoch of 83. This shorter budget was chosen for dataset-scale throughput; it is not assumed equivalent to the reference 3,600-iteration GeoStyle configuration (resolution 1024, batch size four). The final snapshot also retains a small number of earlier longer-budget variants with their original provenance. We additionally consider an optional balanced Jacobian–normal–Laplacian (JN-L) refinement of a coarse deformation proposal, optimized for 120 Adam iterations at learning rate 0.003. The refinement combines Jacobian consistency, source-crease-aware normal preservation, and an area-weighted residual umbrella Laplacian. For coarse vertex positions $\bar{V}$ and residual displacements $r_i = v_i - \bar{v}_i$, the latter term is

$$\mathcal{L}_{\mathrm{L}} = \sum_i m_i \left\| r_i - \sum_{j \in \mathcal{N}(i)} w_{ij} r_j \right\|_2^2, \tag{9}$$

where m_i is an area weight and w_{ij} denotes row-normalized, inverse-edge-length weights on the source mesh. Boundary, nonmanifold, and sharp-crease edges are excluded from this neighborhood. The Laplacian weight is fixed from the initial gradient-norm ratio to the Jacobian and normal terms, rather than tuned using final evaluation scores. Both the original part-aware (PA) proposal and its refined version are considered independently.

Screening and paired observations. Each candidate is assessed against the original source M_0, rather than against its donor or initial proposal. We check canonical correspondence and finite geometry, meaningful deformation, severe assembly failures, paired RGB usability, survival under native 3D representations, and compatibility with the training loader. Multi-view, model-assisted screening also identifies conspicuous spikes, detached components, and assembly collapse; it is not equivalent to exhaustive human review or a guarantee of intersection-free meshes. All accepted pairs retain the source-defined coordinate frame (extent 0.48), original source materials, and matched calibrated

cameras, without independently normalizing the edited geometry. For retargeting, an edited M_1 target-view render specifies the desired state, whereas undeformed M_0 source-view renders supply appearance evidence; neither donor imagery nor ground-truth correspondence is passed to the generator at inference. Approved distinct variants are retained, while near-duplicate final geometries are merged. The frozen snapshot contains 5,611 PA, 4,548 balanced JNL, and six earlier bootstrap variants.

Dataset scope and splits. The resulting dataset contains 10,165 accepted edits from 3,483 source assets spanning 50 categories. A fixed approximately 90/5/5 train/validation/test split contains 9,149/508/508 edits, with all variants of each source UID and existing source-geometry cluster assigned to a single split. The split is disjoint in source identities and geometry clusters, but donor identities are not constrained to be split-disjoint; overlap with earlier multi-view pretraining data has not been established. The dataset is intended to provide plausible and observable shape edits with source-material correspondence, rather than physically certified assemblies or an estimate of the success rate of the underlying deformation procedure.

B. Semantic Feature Matching Details

Descriptors and candidate retrieval. For each observed target patch m, we retrieve the top-64 candidate patches $\mathcal{C}_m$ across the three undeformed source views using cosine similarity of semantic descriptors. We use 1024-dimensional layer-12 FRAME descriptors from a frozen VGGT-Ω backbone [33] for correspondence estimation. These matching descriptors are distinct from the native DINOv3 features subsequently aggregated for 3D generation (Sec. C). Candidate retrieval defines the support over which matching scores are refined; the decoder cannot recover a ground-truth match absent from this initial proposal set.

Matching decoder. We use a lightweight D1 decoder to refine the candidate ranking under geometric deformation. It contains one query-only cross-attention block with four heads of width 32 and a 128-dimensional feed-forward bottleneck. The decoder contextualizes the target query, while the retrieved candidate descriptors remain unchanged. A LayerNorm–MLP scalar head with hidden dimension 64 adds a bounded $0.25\tanh$ residual to the contextualized cosine score. Thus, the final score s_{mk} combines descriptor similarity with a learned, candidate-specific correction; the refined scores are used to construct the matching probabilities in Sec. 3.4.

Correspondence supervision. As described in Sec. A, each source–target mesh pair preserves its ordered face indices and barycentric surface correspondence. We use these correspondences together with training-only depth checks to identify valid positive source patches $\mathcal{P}_m$ for each target patch. The multi-positive listwise objective in Sec. 3.3 assigns probability to the intersection $\mathcal{C}_m \cap \mathcal{P}_m$, allowing more than one valid positive within the retrieved support. Only queries with at least one valid positive candidate contribute to the loss; losses are averaged over eligible queries within each asset, and assets without eligible queries do not update the matcher. Queries without an observable correspondence and cases where the proposal set misses all positives are recorded separately rather than treated as reliable negatives. Ground-truth surface correspondence and depth checks are used for supervision and diagnostic scoring, not for inference-time source selection or visibility gating.

Training and freezing. Of the 9,149 geometry-stage training examples, 9,143 are certified for matcher training; six teacher-rejected cases are excluded from this step only. We train the D1 decoder for seven passes (62,545 updates) using AdamW with learning rate 10^{-4}, weight decay 0.01, and gradient clipping at 1. The VGGT-Ω backbone is frozen during decoder training, and the complete matcher is frozen during subsequent 3D generative fine-tuning. No matching-head training or calibration is performed on the test split.

Matching diagnostics. On 50,493 observable source-camera-ray/target-patch relations from 32 development source clusters, raw cosine matching on the mean-ray support achieves 1.25% exact Top-1 accuracy. Contextualized decoder similarity increases this to 7.73%, and scalar score refinement further increases it to 8.56%. Exact Top-8 coverage increases from 11.09% to 28.47% and 29.92%, respectively. Exact Top-64 coverage remains 50.76% for these variants because they rerank the same proposal support; the additional coverage from combining observed target-member proposals is analyzed separately in Sec. C. These retrieval diagnostics measure correspondence ranking and candidate coverage, not downstream 3D generation quality.

C. Matching-Aware Feature Aggregation Details

Ray grouping and observation support. The geometry (SS) stage uses a 16^3 conditioning grid and 32×32 image-patch maps; SS64 denotes the decoded occupancy resolution rather than the conditioning-grid size. For each source camera j and source patch p, we group conditioning sites that project to the same source-camera patch address into a ray $r = (j, p)$. The corresponding set $\mathcal{M}_r$ contains the *distinct observed target patches* associated with those grid sites. Repeated projections to the same target patch contribute only once, avoiding an implicit preference for

patches intersected by more grid sites. Ray membership and observation support follow the calibrated cameras and image alpha; neither target surface depth nor ground-truth visibility is required at inference.

Ray-level matching posterior. As described in Sec. 3.3, each observed target patch m retrieves up to 64 candidates $\mathcal{C}_m$ from the three source images, using frozen semantic matching descriptors and the D1 decoder's refined scores s_{mk}. We normalize these scores over $\mathcal{C}_m$ with temperature $\tau = 0.05$. A source candidate k indexes both its source view and image patch; candidates absent from a member's proposal set receive zero probability for that member. We average the normalized distributions across $\mathcal{M}_r$, retain the 64 highest-probability candidates from the resulting ray-level mixture, and renormalize their weights, as defined in Eqs. (4)–(5). Unlike matching a single averaged target descriptor, this procedure retains alternative correspondence hypotheses contributed by different target patches before truncation.

Aggregation of native source features. The matching descriptors and the values transferred to the generator are distinct. Matching uses 1024-dimensional layer-12 FRAME descriptors from the frozen VGGT-Ω backbone, whereas the geometry stage gathers 1024-dimensional *native DINOv3 features* at the selected source patches. The final aggregation is the weighted sum in Eq. (6), using the ray-level matching probabilities as weights. The selected Fixed Top-K design uses these native values directly, without an additional learned query/key attention module, value projection, or output projection. Consequently, all conditioning sites in a source-camera ray receive the same retrieved evidence, which is subsequently used to correct the source contribution of the native projection conditioner (Sec. 3.5).

Unsupported observations and fallback. Rays with no observed target members or insufficient valid proposals abstain from cross-state retrieval, leaving the original source-conditioning path unchanged. Importantly, image alpha and camera projection specify observation support rather than physical source–target co-visibility. Likewise, a concentrated matching posterior is not necessarily a calibrated estimate of correspondence correctness. The fallback therefore handles absent evidence but cannot eliminate confidently incorrect matches. Ground-truth surface correspondences and depth checks are used for matcher supervision or diagnostics only, never as inference-time visibility masks.

Aggregation alternatives and diagnostics. Our comparisons include retrieval from the actual Top-1 candidate and uniform weighting over the *same* retained Top-K support. The latter isolates the role of matching-score weights, whereas Top-1 changes both the number of retrieved candidates and their aggregation. We also evaluate learned score refinement as an alternative to fixed weighting. To preserve the initial matching distribution, its per-head attention weights are anchored to the frozen prior:

$$a_{rhk}(\theta) = \mathrm{softmax}_{k\in\mathcal{K}_r}\left[\log p_r(k) + s_{rhk}(\theta) - s_{rhk}(\theta_0)\right], \tag{10}$$

where h indexes four heads and θ_0 denotes initialization. The head probabilities are averaged before aggregating native values; at $\theta = \theta_0$, the weights recover the Fixed posterior. This learned refinement is an evaluated alternative, not part of the selected Fixed Top-K pathway.

On 50,493 observable relations from 32 development source clusters, combining target-member proposals increases exact Top-64 matching coverage from 50.76% to 73.36%. Among 31,408 observable relations where Top-1 misses but Top-K contains a ground-truth candidate, the mean probability assigned to valid candidates is 1.63% for post-hoc uniform weighting, 6.46% for Fixed weighting, and 6.18% for learned refinement. These quantities characterize candidate availability and probability allocation, rather than downstream generation quality; the post-hoc uniform diagnostic is distinct from the separately trained uniform-weighting control.

D. Target-Aware Feature Retargeting Details

Retargeting within the native conditioner. The geometry stage retains the pretrained multi-view projection conditioner, including its target-view contribution and global conditioning tokens. For a source-camera ray r, let $\mathbf{E}_r$ be the matching-weighted source evidence from Sec. 3.4 and $\mathbf{S}_r$ the native source feature already represented in the conditioning pathway. Rather than injecting $\mathbf{E}_r$ as an independent feature, we use the difference $\mathbf{E}_r - \mathbf{S}_r$ to express a correction relative to the existing source contribution. This avoids treating the retrieved evidence as an additional observation and retains the scale and structure of the pretrained conditioning pathway. The correspondence decision is shared by conditioning-grid sites belonging to the same source-camera ray, as described in Sec. C.

Target context and adaptive source use. The retargeter combines the source-feature difference with a native target context $\mathbf{T}_r$ pooled over the conditioning-grid sites of ray r. The target context is distinct from the set of observed target patches used to construct the matching posterior: the latter determines which source candidates are retrieved, whereas $\mathbf{T}_r$ provides target-state information for adapting their features. A learned source-use probability u_r controls the contribution of the correction, together with a bounded signed

gate g on the direct feature difference. The learnable retargeter $\mathcal{R}_\phi$ produces an additional feature residual conditioned on $[\mathbf{E}_r - \mathbf{S}_r, \mathbf{T}_r]$, yielding the correction in Eq. (7). The source-use probability is a conditioning weight, not a calibrated estimate of physical surface visibility or matching correctness.

Residual injection and initialization. Source corrections are organized in their respective source-view patch maps and bilinearly projected into the 3D conditioning grid using the original calibrated camera operators. Their sum is normalized by the total number of conditioning views, consistent with the native projection-based aggregation. The target-view path and global conditioning tokens are not replaced by the retargeted source evidence. We initialize the signed gate to zero and the retargeter output to zero, so the correction initially vanishes and the pretrained conditioning function is recovered. If a ray has no usable matching support, its correction is omitted and the native source contribution is retained rather than replaced with a zero feature. This fallback prevents unsupported updates, although it does not detect confidently incorrect correspondences. Preserving the target conditioning path does not guarantee pixel-exact reproduction of the edited target image after 3D generation.

Stage-specific source adaptation. We adapt the geometry and appearance stages separately rather than imposing one correspondence aggregation strategy on the complete cascade. The sparse-structure stage (SS64) uses the Fixed Top-K evidence of Sec. 3.4 followed by the gated residual retargeter. The intermediate Shape512 and Shape1024 stages remain frozen. Appearance synthesis (PBR1024) operates on the generated geometry and uses its own source-conditioning pathway with a separately trained Top-1 retargeter and native DINOv3/NAF features. Thus, the selected PBR stage does not reuse the SS-stage weighted Top-K aggregation; its Top-1 design is selected from fixed-geometry appearance comparisons.

Fine-tuning and controlled comparisons. We continue the supplied multi-view base-plus-LoRA checkpoints, using the original SS64/Shape512 checkpoint at step $4{,}210$ and Shape1024/PBR1024 checkpoint at step $16{,}850$. Image encoders, the VGGT-Ω matching backbone and decoder, and the intermediate shape stages remain frozen during generative fine-tuning. The geometry stage is trained for 480 updates with global batch size 96, LoRA rank/alpha $16/16$, and denoiser/conditioner learning rates $10^{-5}/10^{-4}$. The appearance stage uses $1{,}900$ updates, global batch size 24, and denoiser/conditioner learning rates $5 \times 10^{-6}/10^{-4}$. Geometry-stage alternatives are compared with common frozen downstream shape models and the same Top-1-retargeting PBR model; appearance-stage alternatives are compared on identical generated geometry, guides, texture noise, rendering triangles, and references. Modules removed in ablations are physically omitted rather than replaced with zero-filled inputs. These controls separate changes in geometry conditioning from changes in appearance synthesis.